

If an AP has first term = a and common difference = d , then the general form of an AP is

$$a, a + d, a + 2d, a + 3d, \dots, a + (n-1)d$$

- n th term of an AP is $T_n = a + (n-1)d$.
- Sum of the first n terms of an AP is

$$S_n = \frac{n}{2}[2a + (n-1)d] \text{ or } S_n = \frac{n}{2}[a + l]$$

where, l = last term

EXAMPLE 1. Find the sum of 11 terms of $-7, -2, 3, 8, \dots$

- a. 200 b. 198 c. 326 d. 137

Sol. d. Here, $a = -7$, $l = -7 + (10 \times 5) = 43$

$$\begin{aligned} \therefore S_{11} &= n \left(\frac{a+l}{2} \right) = 11 \times \left(\frac{-7+43}{2} \right) \\ &= 11 \times \frac{36}{2} = 198 \end{aligned}$$

Arithmetic Mean (AM)

When three terms are in AP, then the middle term is called arithmetic mean of the other two.

If a, b and c are in AP, then b is AM of a and c .

Let a and b be two numbers and M be the AM between them. Then, a, M, b are in AP, $M = \frac{a+b}{2}$.

EXAMPLE 2. Find the arithmetic mean (AM) between 3 and 9.

- a. 4 b. 6
c. 8 d. None of these

Sol. b. The arithmetic mean is $\frac{3+9}{2} = \frac{12}{2} = 6$.

So, 3, 6, 9 are in AP.

Geometric Progression

A geometric progression is a progression of numbers, whose first term is non-zero and each of the term is obtained by multiplying its preceding term by a constant quantity. This constant quantity is called the common ratio of the GP.

Thus, if t_1, t_2 and t_3 are in GP, then common ratio

$$r = \frac{\text{Second term}}{\text{First term}} = \frac{t_2}{t_1}$$

If 'a' is the first term and 'r' is the common ratio, then GP can be written as $a, ar, ar^2, ar^3, \dots, ar^{n-1} = (a \neq 0)$

- n th term of a GP is $T_n = ar^{n-1} = l$.
[where, l = last term]

- Sum of the n terms of a GP is $S_n = \frac{a(r^n - 1)}{r - 1}$,

$$\text{if } r > 1 \text{ and } S_n = \frac{a(1-r^n)}{1-r}, \text{ if } r < 1.$$

- Sum of infinite terms of a GP is $S_n = \frac{a}{1-r}$.

Geometric Mean (GM)

If three terms are in GP, then the middle term is called the geometric mean of the other two. If a, b and c are in GP, then b is the GM of a and c .

Let a and b be two numbers and G be the GM between them. Then, a, G, b are in GP, $G = \sqrt{ab}$, $a > 0$, $b > 0$.

EXAMPLE 3. If the 4th, 10th and 16th terms of a GP are x, y and z respectively, then x, y, z are in

- a. AP b. GP
c. AGP d. HP

Sol. b. Let the first term and common ratio be a and r , respectively.

$$\text{Given, } T_4 = x \Rightarrow ar^{4-1} = x \Rightarrow ar^3 = x \quad \dots(i)$$

$$T_{10} = y \Rightarrow ar^{10-1} = y \Rightarrow ar^9 = y \quad \dots(ii)$$

$$T_{16} = z \Rightarrow ar^{16-1} = z \Rightarrow ar^{15} = z \quad \dots(iii)$$

On multiplying Eq. (i), by Eq. (iii), we get

$$ar^3 \times ar^{15} = xz \Rightarrow a^2 r^{18} = xz$$

$$\Rightarrow (ar^9)^2 = xz \Rightarrow y^2 = xz \quad [\text{from Eq. (ii)}]$$

Hence, x, y and z are in GP.

Harmonic Progression (HP)

A sequence is said to be harmonic progression (HP). If the reciprocals of its terms are in arithmetic progression (AP).

e.g. The sequence, $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$ is an HP because the sequence $1, 3, 5, 7, \dots$ is an AP.

If $a_1, a_2, a_3, \dots, a_n$ are in HP, then $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP.

- n th term of an HP is $T_n = \frac{1}{\frac{1}{a_1} + (n-1)\left(\frac{1}{a_2} - \frac{1}{a_1}\right)}$

Harmonic Mean (HM)

If three terms are in HP, then the middle term is called the harmonic mean of the other two. If a, b and c are in HM, then b is the HM of a and c .

Let a and b be two numbers and H be the HM between them.

Then, a , b and c are in HP

$$\therefore H = \frac{2ab}{a+b}$$

Relation between Arithmetic, Geometric and Harmonic Mean

Let A , G and H be the arithmetic, geometric and harmonic means between a and b , then

$$(i) A \geq G \geq H \quad (ii) G^2 = AH$$

EXAMPLE 4. If a, b, c are in AP, p, q, r are in HP and ap, bq, cr are in GP, then $\frac{p}{r} + \frac{r}{p}$ is equal to

a. $\frac{a}{c} - \frac{c}{a}$
d. $\frac{b}{q} + \frac{q}{b}$

b. $\frac{a}{c} + \frac{c}{a}$
c. $\frac{b}{q} - \frac{q}{b}$

Sol. b. Since, a, b, c are in AP $\Rightarrow b = \frac{a+c}{2}$... (i)

$$p, q, r \text{ are in HP} \Rightarrow q = \frac{2pr}{p+r} \quad \dots (ii)$$

and ap, bq, cr are in GP, $b^2q^2 = apcr$

$$\Rightarrow \left(\frac{a+c}{2}\right)^2 \left(\frac{4p^2r^2}{(p+r)^2}\right) = apcr \quad [\text{from Eqs. (i) and (ii)}]$$

$$\Rightarrow \frac{(a+c)^2 pr}{(p+r)^2} = ac$$

$$\Rightarrow \frac{(a+c)^2}{ac} = \frac{(p+r)^2}{pr}$$

$$\Rightarrow \frac{p^2}{pr} + \frac{r^2}{pr} + 2 = \frac{a^2}{ac} + \frac{c^2}{ac} + 2 \Rightarrow \frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$$

Sum to n Terms of Special Series

The sum of first n terms of some special series is given below

1. The sum of first n natural numbers

$$= \sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

2. The sum of square of the first n natural numbers

$$= \sum n^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. The sum of cubes of the first n natural numbers

$$= \sum n^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

> PRACTICE EXERCISE

1. If $\frac{3+5+7+\dots+n}{5+8+11+\dots+10 \text{ terms}} = 7$, then the value of n

is

- (a) 35 (b) 36 (c) 37 (d) 40

2. If sum of n terms of an AP is $3n^2 + 5n$ and $T_m = 164$, then m is equal to

- (a) 26 (b) 27 (c) 28 (d) None of these

3. In a GP, if the $(m+n)$ th term be p and $(m-n)$ th term be q , then its m th term is

- (a) $\sqrt{pq}$ (b) $\sqrt{p/q}$ (c) $\sqrt{p/p}$ (d) $\sqrt{p+q}$

4. The sum of the first ' n ' terms of the series $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$ is

- (a) $2^n - n - 1$ (b) $1 - 2^{-n}$
(c) $n + 2^{-n} - 1$ (d) $2^n - 1$

5. An AP consists of n (odd terms) and its middle term is m . Then, the sum of the AP is

- (a) $2mn$ (b) $\frac{1}{2}mn$ (c) mn (d) mn^2

6. The sum of $1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \infty$ upto n terms is

- (a) $\frac{25}{16}$ (b) $\frac{15}{16}$
(c) $\frac{5}{16}$ (d) $\frac{3}{2}$

7. The sum of n terms of an AP is $an(n-1)$. The sum of the squares of these terms is equal to

- (a) $a^2n^2(n-1)^2$ (b) $\frac{a^2}{6}n(n-1)(2n-1)$
(c) $\frac{2a^2}{3}n(n-1)(2n-1)$ (d) $\frac{2a^2}{3}n(n+1)(2n+1)$

8. If S be the sum to infinity of a GP, whose first term is a , then the sum of first n terms is
- (a) $S \left(1 - \frac{a}{S}\right)^n$
 (b) $S \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$
 (c) $a \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$
 (d) None of the above
9. If the non-zero numbers a, b, c are in AP and $\tan^{-1} a, \tan^{-1} b, \tan^{-1} c$ are also in AP, then
- (a) $a = b = c$ (b) $b^2 = 2ac$
 (c) $a^2 = bc$ (d) $c^2 = ab$
10. $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$, then a, b, c are in
- (a) AP (b) GP
 (c) HP (d) None of these
11. The value of $x + y + z$ is 15, if a, x, y, z, b are in AP while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{3}$, if a, x, y, z, b are in HP. Then, a and b are
- (a) 1, 9 (b) 3, 7
 (c) 7, 3 (d) None of these
12. If the m th and n th term of a HP are n and m respectively, then the mn th term is
- (a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$
13. If $a, 2a + 2, 3a + 3$ are in GP, then what is the fourth term of the GP?
- (a) -13.5 (b) 13.5 (c) -27 (d) 27
14. If the sum of first ' n ' natural numbers is $\frac{n(n+1)}{2}$. Then, what will be the sum of first ' n ' terms of the series of alternate positive and negative numbers when ' n ' is even?
- $$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \dots$$
- I. $\frac{n(n+1)}{2}$ II. $\frac{n^2(n+1)}{2}$ III. $\frac{-n(n+1)}{2}$
- Which of the above statement(s) is/are correct?
- (a) Only I (b) Only III (c) Only II (d) None of these

> PREVIOUS YEARS QUESTIONS

15. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, then which one of the following relations is correct?
- (a) G is the geometric mean between A and H **2015 I**
 (b) A is the arithmetic mean between G and H
 (c) H is the harmonic mean between A and G
 (d) None of the above
16. Consider the following statements in respect of the expression $S_n = \frac{n(n+1)}{2}$, where ' n ' is an integer.
- I. There are exactly two values of n for which $S_n = 861$.
 II. $S_n = S_{-(n+1)}$ and hence for any integer m we have two values of n for which $S_n = m$.
- Which of these statement(s) is/are correct? **2016 (I)**
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

> ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16									

> HINTS AND SOLUTIONS

1. (a) $\therefore \frac{\frac{n}{2}[2 \times 3 + (n-1)2]}{\frac{10}{2}[2 \times 5 + (10-1) \times 3]} = 7$

$$\Rightarrow \frac{n(n+2)}{5 \times 37} = 7$$

$$\Rightarrow n^2 + 2n - 1295 = 0$$

$$\Rightarrow n^2 + 37n - 35n - 1295 = 0$$

$$\Rightarrow (n+37)(n-35) = 0$$

$$\therefore n = 35$$

2. (b) $\therefore T_m = S_m - S_{m-1}$

$$\Rightarrow 164 = 3(2m-1) + 5 \cdot 1$$

$$\Rightarrow 6m = 162$$

$$\therefore m = 27$$

3. (a) $T_{m+n} = ar^{m+n-1} = p$,
 $T_{m-n} = ar^{m-n-1} = q$
 On multiplying, we get $a^2 r^{2m-2} = pq$
 $\therefore T_m = ar^{m-1} = \sqrt{pq}$

4. (c) $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots + n$ terms

$$= \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{4}\right) + \left(1 - \frac{1}{8}\right) + \dots + \left(1 - \frac{1}{2^n}\right)$$

$$= n - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}\right)$$

$$= n - \frac{1}{2} \left(\frac{1 - (1/2)^n}{1 - 1/2}\right) = n + 2^{-n} - 1$$

5. (c) Middle term = $T_{\frac{n+1}{2}}$
 $\therefore a + \left(\frac{n+1}{2} - 1\right)d = m$ [given]
 $2a + (n-1)d = 2m \quad \dots(i)$
 Now, $S_n = \frac{n}{2}[2a + (n-1)d] = nm$

6. (a) The given sequence is arithmetic geometric series, where $r = \frac{1}{5}$ and $d = 1$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} + \frac{dr}{(1-r)^2} \\ &= \frac{1}{1-\frac{1}{5}} + \frac{1 \times \frac{1}{5}}{\left(1-\frac{1}{5}\right)^2} \\ &= \frac{5}{4} + \frac{5}{16} = \frac{25}{16} \end{aligned}$$

7. (c) Let $S_n = an(n-1)$, then
 $S_{n-1} = a(n-1)(n-2)$
 $\therefore T_n = S_n - S_{n-1} = 2a(n-1)$
 $T_{n-2} = 4a^2(n-1)^2$
 $\therefore \text{Sum} = \Sigma T_n^2 = 4a^2 \frac{(n-1)(n)(2n-1)}{6}$
 $= \frac{2a^2 n(n-1)(2n-1)}{3}$

8. (b) Let r be the common ratio of GP, then

$$\begin{aligned} S &= \frac{a}{1-r}, \quad r = 1 - \frac{a}{S} \\ \therefore S_n &= \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} (1-r^n) \\ &= S \left[1 - \left(1 - \frac{a}{S}\right)^n \right] \end{aligned}$$

9. (a) Since, $2b = a + c \quad \dots(i)$
 and $2 \tan^{-1} b = \tan^{-1} a + \tan^{-1} c$
 $\Rightarrow \frac{2b}{1-b^2} = \frac{a+c}{1-ac}$
 $\Rightarrow b^2 = ac$ [from Eq. (i)]
 $\Rightarrow 4b^2 = 4ac$
 $\Rightarrow (a+c)^2 - 4ac = 0$ [from Eq. (i)]
 $\Rightarrow (a-c)^2 = 0$
 $\Rightarrow a = c = b$

10. (c) Given, $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$
 $\Rightarrow \frac{1}{b-a} - \frac{1}{c} = \frac{1}{a} - \frac{1}{b-c}$

$$\begin{aligned} \Rightarrow \frac{(c-b+a)}{d(b-a)} &= \frac{(b-c-a)}{a(b-c)} \\ \Rightarrow \frac{1}{d(b-a)} &= -\frac{1}{a(b-c)} \\ \Rightarrow ba - ca &= -cb + ac \\ \Rightarrow ab + bc &= 2ac \\ \therefore b &= \frac{2ac}{a+c} \end{aligned}$$

Hence, a, b, c are in HP.

11. (a) $\therefore a, x, y, z, b$ are in AP.

$$\begin{aligned} \therefore x + y + z &= 3\left(\frac{a+b}{2}\right) \\ \Rightarrow 15 &= 3\left(\frac{a+b}{2}\right) \\ \Rightarrow a + b &= 10 \quad \dots(i) \\ \text{Also, } a, x, y, z, b &\text{ are in HP.} \\ \Rightarrow \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b} &\text{ are in AP.} \\ \Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 3\left(\frac{a+b}{2ab}\right) \\ \Rightarrow \frac{5}{3} &= \frac{3 \times 10}{2ab} \quad [\because a+b=10] \\ \Rightarrow ab &= 9 \end{aligned}$$

On solving Eqs. (i) and (ii), we get
 $a = 1, b = 9$ or $b = 1, a = 9$

12. (b) Let a be the first term and d the common difference of corresponding AP.

So, m th and n th term of AP are $\frac{1}{n}$ and $\frac{1}{m}$.

$$\therefore \frac{1}{n} = a + (m-1)d \quad \dots(i)$$

$$\text{and } \frac{1}{m} = a + (n-1)d \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get,

$$d = \frac{1}{mn} \text{ and } a = \frac{1}{mn}$$

$\therefore m$ th term of AP

$$\begin{aligned} &= \frac{1}{mn} + (mn-1) \times \frac{1}{mn} \\ &= \frac{1+mn-1}{mn} = 1 \end{aligned}$$

$\therefore m$ th term of HP = 1

13. (c) $a, 2a+2, 3a+3$ are in GP.
 $\Rightarrow (2a+2)^2 = (3a+3)a$
 $\Rightarrow 4a^2 + 4 + 8a = 3a^2 + 3a$
 $\Rightarrow a^2 + 5a + 4 = 0$
 $\Rightarrow a = -1, -4$

Now, $a = -1$ does not satisfy the given series.

$\therefore -4, -6, -9$ are in GP.

$$\therefore t_4 = -4\left(\frac{3}{2}\right)^3 = -13.5$$

14. (b) When ' n ' is even.

$$\begin{aligned} \text{Let } n &= 2m, \text{ then} \\ &= 1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \dots \\ &= (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2) \\ &\quad + \dots + (2m-1)^2 - (2m)^2 \\ &= (1+2)(1-1) + (3+4)(-1) \\ &\quad + (5+6)(-1) + \dots \\ &\quad + (2m-1+2m)(-1) \\ &= -(1+2+3+4+\dots+2m) \\ &= \frac{-2m(2m+1)}{2} = \frac{-n(n+1)}{2} \end{aligned}$$

15. (a) Given, A, G and H are the arithmetic, geometric and harmonic means between a and b , respectively.

$$\therefore A = \frac{a+b}{2}, \quad G = \sqrt{ab} \text{ and } H = \frac{2ab}{a+b}$$

(i) (ii) (iii)

On multiplying Eq. (i) and (iii), we get

$$\therefore AH = \frac{a+b}{2} \times \frac{2ab}{a+b} = ab = (\sqrt{ab})^2$$

[from Eq. (ii)]

$$AH = G^2$$

Hence, the option (a) is correct.

16. (a) I. $S_n = \frac{n(n+1)}{2} = 861$

$$\Rightarrow n^2 + n - 861 \times 2 = 0$$

$$\Rightarrow (n+42)(n-41) = 0$$

$$\Rightarrow n = -42, 41$$

Hence, statement I is correct.

- II. Given, $S_n = S_{-(n+1)}$
 If $S_n = m$, then we have two values of n if and only if m is positive integer.

Hence, statement II is incorrect.

03

HCF AND LCM OF NUMBERS

Usually (2-3) questions have been asked from this chapter. Generally questions based on LCM and HCF are related to traffic lights, racetracks, largest size of tile etc. This concept is also useful in the chapters of time and distance, time and work, pipes and cisterns etc.



FACTORS

If number ' a ' divides the number b without leaving a remainder, then ' a ' is said to be the factor of ' b '.

e.g. (i) 4 is a factor of 16 as $16 = 4 \times 4$ (ii) 9 is a factor of 729 as $729 = 9 \times 9 \times 9$

Prime Factors

The factors that cannot be again factorized i.e. the factors which are prime numbers are called prime factors.

e.g. (i) 9 is a factor of 729, but 3 is a prime factor of 729.

(ii) 8 and 9 are factors of 72, but 2 and 3 are prime factors of 72.

Least Common Multiple (LCM)

1. **Common Multiple** A common multiple of two or more numbers is a number which is exactly divisible (without leaving remainder) by each of them.

e.g. 30 is a common multiple of 2, 3, 5, 6, 10 and 15 because 30 is exactly divisible by each number.

Similarly, 72 is a common multiple of 2, 3, 4, 6, 9, 12 and 18.

2. **Least Common Multiple** The least common multiple of two or more given numbers is the least number which is exactly divisible by all the given numbers.

e.g. 84, 162, 252 are the common multiples of 2, 3, 4, 7. But, 84 is the LCM of 2, 3, 4, 7.

Methods of Finding LCM

1. **Prime Factorization Method** Write down the given numbers as the product of prime factors. Then, the LCM is the product of the highest powers of all the prime factors.